Susceptibility and Spatial Signature of Abrupt Thaw Across Alaska's Permafrost Landscapes

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Key points

- Key point 1
- Key point 2
- Key point 3

Abstract

Abrupt permafrost thaw disproportionately accelerates carbon release and damages infrastructure, yet existing spatial products primarily describe thaw occurrence, landform, or stage rather than thaw mode. We train a gradient-boosted classifier on $19,288$ expert-labeled observations, using 70 environmental variables to distinguish abrupt from non-abrupt thaw across Alaska. We evaluate the model's generalizability using spatial-block cross-validation, convert model output into a prior-free log-evidence index, bound our predictions with an area-of-applicability (AoA) analysis, and use grouped Shapley (SHAP) values to identify key indicators. The classifier achieves an area under the precision-recall curve (AUC-PR) of 0.852 ± 0.011 for the minority non-abrupt class under 10-km spatial cross-validation, compared with a prevalence floor of 0.057. Within the AoA, which covers about 90% of Alaska's permafrost domain, we find 24.7% of the landscape has environmental features more consistent with abrupt than non-abrupt thaw, corresponding to approximately $264,000$ km^2. Abrupt-favoring terrain is concentrated in the northern and western tundra regions, including roughly two-thirds of the Brooks Range and Seward Peninsula. Topography and temperature provide the leading differentiation between thaw modes, and the model links both flat, low-lying wetlands and steep uplands to abrupt thaw, consistent with distinct pathways of thaw and known landforms. Previously mapped thermokarst occurrence classes explain only $1-7\%$ of the variance in log-evidence, showing that thaw mode provides a largely distinct and complementary new dimension of permafrost change. The resulting surface characterizes landscape predisposition to abrupt thaw, and should guide future field campaigns, permafrost-carbon feedback models, and regional hazard assessments.

Plain Language Summary

Permafrost is ground that remains frozen for two consecutive years or more, and it does not thaw uniformly everywhere. Abrupt thaw, marked by extreme and rapid changes to ecosystems and the land surface, exposes deep carbon stores and threatens human infrastructure and communities. Here, we train a machine learning model on $19,288$ observations of permafrost thaw, labeled by experts, using 70 different environmental factors to map where conditions across Alaska resemble those linked to abrupt or non-abrupt thaw. We test the model against points from geographic areas it has not seen in its training data, and limit its predictions to places where the landscape is similar to our training points. Within this area, about one quarter, or $264,000$ km^2, has conditions that are more consistent with abrupt thaw than non-abrupt thaw. These areas concentrate in the northern and western tundra, including much of the Brooks Range and Seward Peninsula. Topography and temperature are the most important sets of indicators for the model to distinguish between abrupt and non-abrupt thaw. The map we build does not show where thaw is happening now, nor predict where it will occur in the future. Instead, it shows us where Alaska's landscapes may be susceptible to abrupt thaw, helping guide field observations, climate modeling, and studies of regional hazards.

1 Introduction

The northern permafrost region contains more organic carbon in its upper three meters of soil than currently resides in Earth's atmosphere (Hugelius et al., 2014). Most permafrost thaw proceeds by gradual, top-down deepening of the active layer, but abrupt thaw destabilizes and collapses the ground surface over days to years, mobilizing carbon stocks that gradual thaw would take decades to reach (Turetsky et al., 2020; Nitzbon et al., 2020). Under high emissions scenarios, abrupt thaw is projected to impact only 2.5 million km^2 out of the 18 million km^2 permafrost region, but threatens to release nearly as much carbon as gradual thaw processes operating over a much larger area (Turetsky et al., 2020). Abrupt thaw leads to a variety of landforms, including thermokarst lakes, collapsing peatlands, and hillslope failures, all of which differ in the quantity and form of carbon released (Olefeldt et al., 2016; Schädel et al., 2016). The scale and timing of carbon release from permafrost soils therefore depend on how and where the landscape undergoes abrupt thaw, and this spatial pattern remains one of the largest sources of uncertainty in the permafrost-carbon feedback (Turetsky et al., 2020; Walter Anthony et al., 2018).

Beyond climate feedbacks, abrupt thaw represents an immediate hazard to the roughly 5 million people who live in the Arctic circumpolar permafrost region (Ramage et al., 2021). Projections suggest that 30-50% of Arctic infrastructure may be at risk of damage by permafrost thaw by 2050, threatening tens of billions of U.S. dollars in damage by the end of the century (Hjort et al., 2022). In Alaska alone, permafrost thaw may cause $37 to $51 billion in damage to buildings and roads (Manos et al., 2025). This burden will fall disproportionately on remote and Indigenous communities, whose unofficial, "vernacular" infrastructure and assets often lie outside conventional damage inventories (Verstraete & Zachwieja, 2025). Between 1974 and 2019, thermokarst at Point Lay, Alaska, expanded nearly an order of magnitude faster beneath the developed village than in the surrounding tundra, indicating a potential positive feedback between the built environment and abrupt thaw (Jones et al., 2025). Predicting how and where abrupt thaw will develop on the landscape is therefore critical for helping Arctic communities adapt to or mitigate the effects of future climate change.

Abrupt permafrost thaw proceeds faster than gradual thaw, by definition, but also results from different processes, leaves distinct surface expressions, and has specific consequences for ecosystems, hydrology, and carbon release (Webb et al., 2025). Gradual thaw typically proceeds from the top down as a progressive deepening of the seasonally thawed active layer into near-surface permafrost (Walter Anthony et al., 2018; Webb et al., 2025). Abrupt thaw, by contrast, is not confined to this vertically ordered progression: subsidence, erosion, mass wasting, and the flow of water and heat can rapidly propagate thaw downward and laterally, affecting deeper or adjacent permafrost (Fig. 1) (Turetsky et al., 2020; Nitzbon et al., 2020; Webb et al., 2025). The resulting changes to the landscape occur quickly relative to geomorphic time scales, from vertical subsidence of almost a meter over a decade (Farquharson et al., 2019) to thaw-slump headwall retreat of more than ten meters in a single year (S. Kokelj et al., 2015). A site's thaw mode, abrupt or gradual, is not dictated by the presence of permafrost alone, but emerges from an interplay of ground ice, geomorphology, hydrology, and disturbance history (Jorgenson & Osterkamp, 2005; S. V. Kokelj & Jorgenson, 2013).

Permafrost has been mapped extensively across the Arctic, but existing products largely resolve conditions other than the mode of thaw. Many characterize the substrate, outlining permafrost extent and zonation (Brown et al., 1997), ground ice content (Jorgenson et al., 2008), and probability of occurrence (Obu et al., 2019). Others document where abrupt thaw features have formed, such as ponds and lakes (Muster et al., 2017) or retrogressive thaw slumps (Nitze et al., 2025), or quantify how far degradation of specific features has advanced (Zhang et al., 2025). A growing set of susceptibility maps predicts where specific abrupt thaw processes may develop, including thermokarst lakes (Wang et al., 2023), retrogressive thaw slumps (Makopoulou et al., 2024), and active layer detachment (Makopoulou et al., 2025), and a recent hemispheric model combines lake and slump susceptibility into a single product (Yang et al., 2025). Each of these efforts estimates the likelihood that a named landform will occur at any given point, and the combined model remains a union of predictions anchored to specific processes. None of these previous efforts addresses the question of thaw mode: whether the conditions at a site resemble those under which abrupt or gradual thaw has been observed, independent of any single landform (Fig. 1). Across Alaska, that contrast is carried only by point-wise samples from expert analysis of field and remote sensing observations (Webb et al., 2026), and not yet mapped as a continuous, predictive surface.

Here, we map thaw mode as a continuous, predictive surface across Alaska. We train a gradient-boosted classifier on 19,288 expert-labeled thaw points from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The classifier evaluates sites along a gradient of how strongly key environmental variables resemble either abrupt or non-abrupt thaw, using 70 geospatial indicators sourced from remote sensing, community data products, and model reanalysis. To mitigate sampling bias inherent to the Thaw Database, we evaluate the model under spatial cross-validation, bounded by an applicability mask that indicates where points fall within the envelope of the training data, and then apply it to a statewide feature grid populated by the same data sources. The resulting product is a prior-free, log-evidence index of thaw mode. We then use grouped Shapley (SHAP) values (Lundberg & Lee, 2017; Lundberg et al., 2020) to identify which geospatial indicators best distinguish between thaw modes and how they shift the classifier's prediction. This paper describes the expert-labeled points and feature stack in Section 2, model construction and evaluation in Section 3, the susceptibility map and its drivers in Section 4, and limitations, interpretations, and comparisons with the broader permafrost mapping literature in Section 5.

2 Data

2.1 Labeled thaw points

We train and evaluate the classifier on expert-labeled observations of thaw mode from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The Thaw Database compiles observations of permafrost across Alaska from a wide variety of sources, including both field and remote-sensing studies. After removing exact duplicates in feature space, the training set comprises 19,288 points, each assigned a binary thaw mode label. We encode abrupt thaw as `class 0`, and then all non-abrupt thaw labels as `class 1`. Abrupt thaw dominates the dataset with 18,181 points (94.26%), alongside only 1,107 (5.74%) non-abrupt thaw points. Notably, this composition is the inverse of the landscape it samples. The majority of observations in the Thaw Database are `class 1`, yet abrupt thaw is projected to impact only a fraction of the permafrost region overall (Turetsky et al., 2020).

The non-abrupt thaw label denotes sites undergoing gradual rather than abrupt thaw. Roughly 18% of these points are drawn from the long-term permafrost monitoring networks (the Global Terrestrial Network for Permafrost boreholes and Circumpolar Active Layer Monitoring program). We label these points as "non-abrupt" rather than

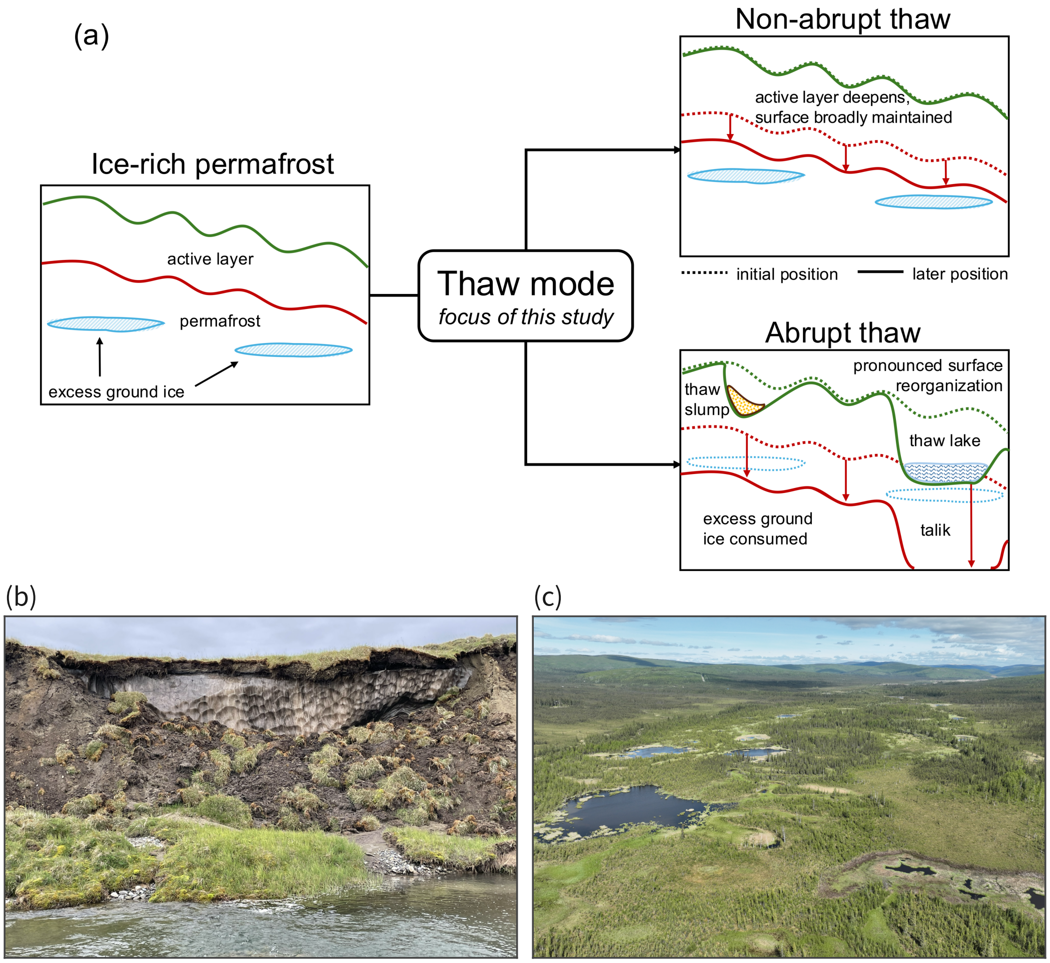

Figure 1. Conceptual illustration of permafrost thaw mode. Schematic diagram of an idealized cross-section (a), showing an ice-rich initial condition, not drawn to scale. Under thaw, the landscape transitions into one of two thaw modes. Non-abrupt thaw (top) broadly maintains the surface, allowing for some amount of gradual subsidence, while abrupt thaw (bottom) is accompanied by pronounced surface reorganization. Field photographs show two examples of abrupt thaw landforms: a thaw slump along an eroding riverbank in Alaska's North Slope (b), photographed by Irina Overeem, and a drone image of a thermokarst wetland in Goldstream Valley (c), photographed by Kevin Rozmiarek.

"gradual" thaw deliberately, following the convention in the Thaw Database, to include sites where thaw is not definitively abrupt. The abrupt thaw class contains a variety of landforms, including thermokarst lakes, collapsing peatlands, retrogressive thaw slumps, and active-layer detachments, among others. We refer the reader to Webb et al. (2026) for the full decision matrix used to label points.

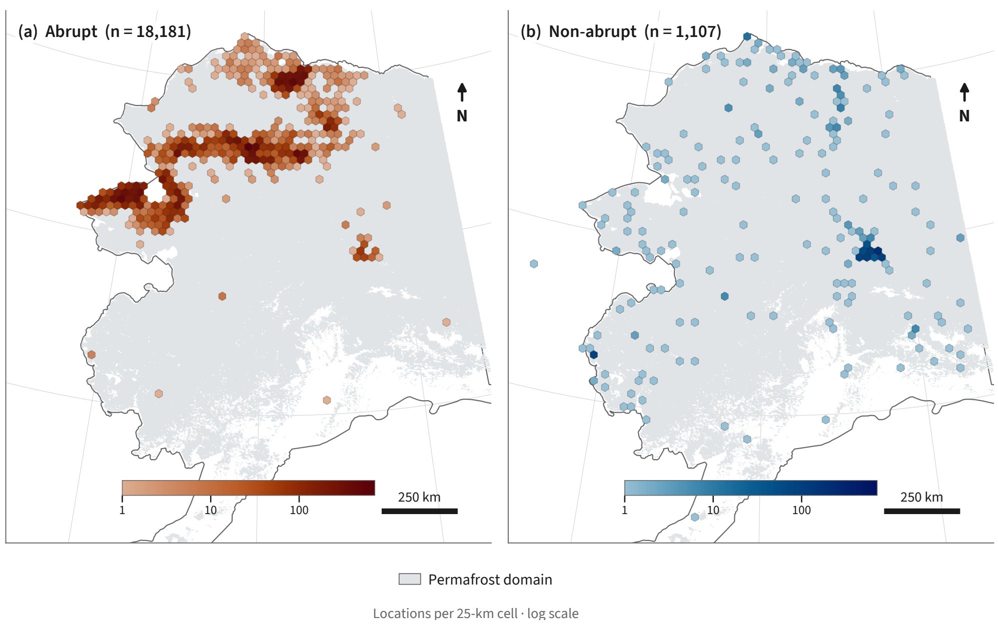

Figure 2. Spatial distribution of thaw mode observations. Locations of (a) 18,181 abrupt thaw points and (b) 1,107 non-abrupt thaw points retained from the Alaska Permafrost Thaw Database (Webb et al., 2026) after pre-processing. Hexagons aggregate observations into 25-km cells, with color intensity indicating the number of observations per cell on a shared logarithmic scale. Grey shading denotes the permafrost domain defined by Obu et al. (2019). The mapped densities reflect the distribution of source studies and sampling effort, not the landscape prevalence of either thaw mode.

Because the observations were largely assembled from field campaigns and targeted remote-sensing surveys, the spatial distribution of labeled points reflects the research focus of the contributing studies and physical accessibility of sites. Figure 2 shows the distribution of abrupt (a) and non-abrupt (b) thaw observations. The two classes are not evenly balanced across the landscape, with many more abrupt thaw studies focused in the north, and non-abrupt sites spread more evenly across the state. This uneven coverage motivates the spatial cross-validation and area-of-applicability analyses described in Section 3. We carry site coordinates through the pipeline as metadata and use them to build cross-validation blocks, but exclude them from the feature table so that the model cannot use location itself as a predictor.

2.2 Geospatial features

Each point is characterized by a common set of geospatial features extracted at its location, which together form a feature table of 19,288 rows and 70 columns. Continuous variables enter the matrix directly, while categorical variables are one-hot encoded into individual columns. Data sources include remote sensing products, modeled or interpolated data products, and gridded climate products (Table A1).

We obtain surface topography from the USGS 3D Elevation Program 10 m digital elevation model (U.S. Geological Survey, 2019). We use elevation, slope, aspect (decomposed into northness and eastness), and mean curvature computed at 500 m and 2 km scales. We account for surface hydrology using MERIT Hydro (Yamazaki et al., 2019), pulling height above nearest drainage and upstream contributing area. Land cover classifications come from the National Land Cover Database for Alaska (Dewitz, 2019), and are one-hot encoded into 18 indicator columns. Fire history is derived from the MODIS MCD64A1 burned-area product (Giglio et al., 2018), expressed as time since last fire and cumulative burn count.

We obtain soil properties from SoilGrids (Poggio et al., 2021), including organic carbon, total nitrogen, bulk density, and soil texture in sand, silt, and clay fractions. To avoid collinear columns (where sand, silt, and clay fraction sum to a fixed total), we encode soil texture as sand fraction and clay fraction, dropping silt from the feature table. We then aggregate the soil data into two depths, "surface" from 0-30 cm and "depth" from 30-200 cm, to aid in later interpretability. We include relative flammability as a continuous column and dominant vegetation mode encoded into 7 indicator columns, both sourced from the ALFRESCO landscape succession model (Bennett et al., 2017). We include a binary classification of yedoma from the Ice-Rich Yedoma Permafrost Inventory (Strauss et al., 2022), denoting points that are classified as "confirmed" yedoma in the IRYP inventory. Nineteen bioclimatic variables describe annual and seasonal temperature and precipitation, sourced from the WorldClim v1 climatology (Hijmans et al., 2005). Mean annual snow water equivalent (SWE) and multidecadal trends in SWE, precipitation, and temperature come from Daymet v4 (Thornton et al., 2022), computed from 1991-2020.

Across all data products, we do not impute missing values, choosing instead to select a classification algorithm that handles NaN values natively (Section 3). Two sources have structured gaps worth noting: SoilGrids is missing at approximately 16% of our training points, and the MODIS fire domain thins considerably above 70°N, leaving the northern Arctic coast without fire history coverage. Table A1 lists all data products with their native resolutions and references.

2.3 Prediction datacube

Once we have a trained classifier, producing predictions of thaw mode across Alaska requires mapping the same 70 geospatial indicators onto a statewide grid. We build the prediction datacube at a nominal resolution of 1 km on a fixed 0.00898° grid in geographic coordinates (EPSG:4326). True cell area varies with latitude, narrowing towards the north. We sample each data product onto the grid with the exact same extraction and post-processing used for the feature table, described above. Continuous rasters are resampled from their native resolution to the grid resolution, categorical layers are sampled by nearest neighbor, and yedoma is assigned by polygon membership. This parity ensures that the trained model transfers from the training points to the statewide grid without a train/serve mismatch. We restrict all predictions to the permafrost domain defined by the Obu et al. (2019) permafrost probability map, where the classification of thaw mode is physically meaningful.

3 Methods

To map thaw mode across Alaska, we train a gradient-boosted classifier on the labeled thaw points, evaluate its skill with spatial cross-validation, and bound it with an area-of-applicability analysis. Our result is a log-evidence index of thaw mode, which we then map statewide by applying the trained model. Our methods account for the spatial clustering of abrupt thaw points, the relative sparsity of non-abrupt thaw points in the training dataset, and the lack of expert-verified thaw observations from vast regions within the state. After training and deploying the model, we then apply post-hoc interpretability tools to understand the features that drive the model and produce the strongest indications of abrupt thaw.

3.1 Gradient-boosted classification

We use gradient-boosted decision trees to classify thaw mode, following the XGBoost algorithm (Chen & Guestrin, 2016). XGBoost has two key advantages for our problem. First, several of our features have missing values that are structured, rather than simply denoting a random point of missing data. For example, SoilGrids does not carry soil data over open water features, and similarly, the MODIS fire data is not available at very high latitudes. Gradient-boosted trees learn directly from incomplete rows, and do not require us to invent a structure for missing values or handle them ad hoc. Second, XGBoost allows for exact per-feature attribution through the TreeSHAP algorithm (Lundberg et al., 2020), which we use to build model interpretability (§3.4).

We measure model skill as the area under the precision-recall curve, as appropriate for a rare positive class (i.e., non-abrupt thaw here) (Saito & Rehmsmeier, 2015). Overall accuracy and the area under the receiver operating characteristic (ROC) curve are both dominated by abrupt thaw, and can appear high even when the model fails to identify non-abrupt sites. We report a single ROC value for comparison with other work, but otherwise evaluate the model through precision-recall. We select hyperparameters to maximize this score (Appendix B) using nested cross-validation and then refit the final model on all labeled points. Rather than reweight the classes (abrupt and non-abrupt) to compensate for their relative prevalence in the database, we keep the sample prior intact, allowing us to later produce a prior-free index of thaw mode (§3.3). We also fit a logistic regression model to the data to act as a baseline model, but we do not test our pipeline against any other machine learning algorithms.

3.2 Spatial evaluation

Figure 2 shows the spatial clustering of abrupt and non-abrupt thaw points from the training dataset. The underlying variables that we use to predict thaw mode vary smoothly in space, following Tobler's first law of geography that "near things are more related than distant things" (Tobler, 1970). Therefore, a random test-train split on these data would place near-duplicate neighbors on either side of the split, inflating the model's apparent skill.

To avoid this bias, we evaluate the model using spatial cross-validation. We divide the state into square blocks on an equal-area grid and hold out entire blocks together, so that clusters of nearby points fall on the same side of the test-train split, rather than across it. The appropriate block size is not obvious, so we repeat the evaluation on blocks ranging from 10 to 200 km on a side. Larger block sizes push the test points further away from the training data, so this systematic approach gives one measure of how well the model generalizes as we increase the distance away from its training data. We do not use an additional buffer zone around the blocks, as a sweep of buffer sizes indicated that this was irrelevant for leakage.

Block cross-validation tests the model's ability to generalize beyond immediate neighborhoods, but does not scale up to a full statewide analysis. To probe statewide generalization, we also withhold entire geographic regions in turn and measure how skill degrades as the model is forced to predict on progressively less sampled terrain. We repeat the block-to-fold assignment many times to confirm that measured skill does not depend on any single partition.

3.3 Mapping thaw mode

Applying the trained model to our statewide feature grid yields an estimated probability that any given cell exhibits abrupt thaw. However, we do not report this probability directly. The model is calibrated to the class balance in the training database, and that balance reflects how the observations were collected much more than the true proportion of abrupt thaw across Alaska. Prior work suggests abrupt thaw will affect a minority of the permafrost region (Turetsky et al., 2020), rather than the roughly 94% that random sampling of our database would imply. The true landscape proportion is not recoverable from a biased sample, so rather than use a calibrated probability, we instead build a prior-free index of evidence. In plain language, a high index value indicates high evidence that a grid cell matches the features associated with abrupt thaw.

For a cell with a feature vector $\vec{x}$, let $p(\vec{x})$ be the model's estimated probability of abrupt thaw, and then let Π be the prevalence of abrupt thaw in the training data. We then define the log-evidence index as:

$$s(\vec{x}) = \text{logit}\big(p(\vec{x})\big) - \text{logit}(\Pi), \qquad (1)$$

where $\text{logit}(z) = \log[z/(1-z)]$ maps a probability to a real number. This is the logarithm of the likelihood ratio between abrupt and non-abrupt thaw, as indicated by the local features at a given grid cell. A value of zero is neutral, while positive values favor abrupt thaw and negative values favor non-abrupt thaw. The index is neither a probability nor a discrete class, so we will call it a *log-evidence index* here.

Because we did not reweight the classes during training, the model holds the sample prevalence Π as its prior, and then subtracting $\text{logit}(\Pi)$ removes it. This means that the resulting index does not depend on class balance at all. Given new observations with a significantly different ratio of abrupt to non-abrupt thaw points, $p(\vec{x})$ would change, but $s(\vec{x})$ would not, because the prior is removed from the index. This insulates the index from raw class balance concerns, but does not yet address sampling that is uneven across feature space.

The model will return a prediction at any grid cell, but the training dataset does not cover every combination of environmental variables found across the state of Alaska. Where a cell lies far away from the training data in feature space, the model extrapolates and its output should not be trusted. To delineate these regions, we adapt the area-of-applicability method from Meyer and Pebesma (2021). For each cell, we calculate a dissimilarity index, measuring the distance in feature space to the nearest training point, with features weighted by importance to the model and scaled by average dissimilarity among training points. We set the applicability threshold at the 99.9th percentile of this index measured across all of the training points, corresponding to a dissimilarity index of 0.27. Cells outside of that range are flagged as extrapolation. This threshold does not denote a limit of model skill, as we found that the model performance did not degrade as the dissimilarity index rose (Appendix C). Instead, the threshold is chosen as the reasonable extent of the feature space, beyond which skill cannot be measured regardless.

3.4 Interpretability

To interpret the model's classification decisions, we aim to attribute output values to input features. Many of the features are strongly correlated with one another, so a simple per-feature interpretation would risk diluting one physical signal across several environmental variables. As such, we group the features in the model first, before attributing importance, using the correlation structure of the features alone. To do so, we measure the correlation between every pair of features and join similar features into families by hierarchical clustering. Our categorical variables, which are encoded as one-hot columns, are collapsed into families by data source. Features that are strongly anti-correlated are treated as redundant as well, because a variable and its inverse carry the same information. The number of families is not fixed in advance, but emerges from the correlation structure (Appendix D).

With the feature families defined, we quantify each group's contribution to the model using SHAP values (Lundberg & Lee, 2017), computed for our tree model with the TreeSHAP algorithm (Lundberg et al., 2020). A SHAP value measures how far a feature moves the model's output for a single prediction, as compared to its average output. We sum the SHAP values within each family to obtain one contribution per family, and keep the same groupings throughout.

We compute two different sets of SHAP values, for two purposes. First, to rank the families by importance, we compute SHAP values on held-out points during spatial cross-validation, refitting the model within each fold so that every attribution is made on data the model did not see in training. This provides an unbiased ranking on held-out data. Second, to map how the drivers of abrupt thaw vary across the state, we compute SHAP values from the final, all-data model at every grid cell and record the family that dominates in each. This gives a spatially complete display, using the actual deployed model. Both computations share the same family definitions, so the ranking and the map remain consistent.

4 Results

4.1 Alaska's susceptibility to abrupt thaw

Figure 3a maps the log-evidence index of thaw mode across Alaska's permafrost domain. Within the area of applicability (AoA), 24.7% of the landscape exhibits features that are more consistent with abrupt thaw than with non-abrupt thaw, corresponding to an area of roughly 264,000 km^2 out of the 1.07 million km^2 of the in-AoA permafrost domain. We denote a log-evidence of zero as the neutral boundary, where features favor neither abrupt nor non-abrupt thaw. The log-evidence map describes the current char-

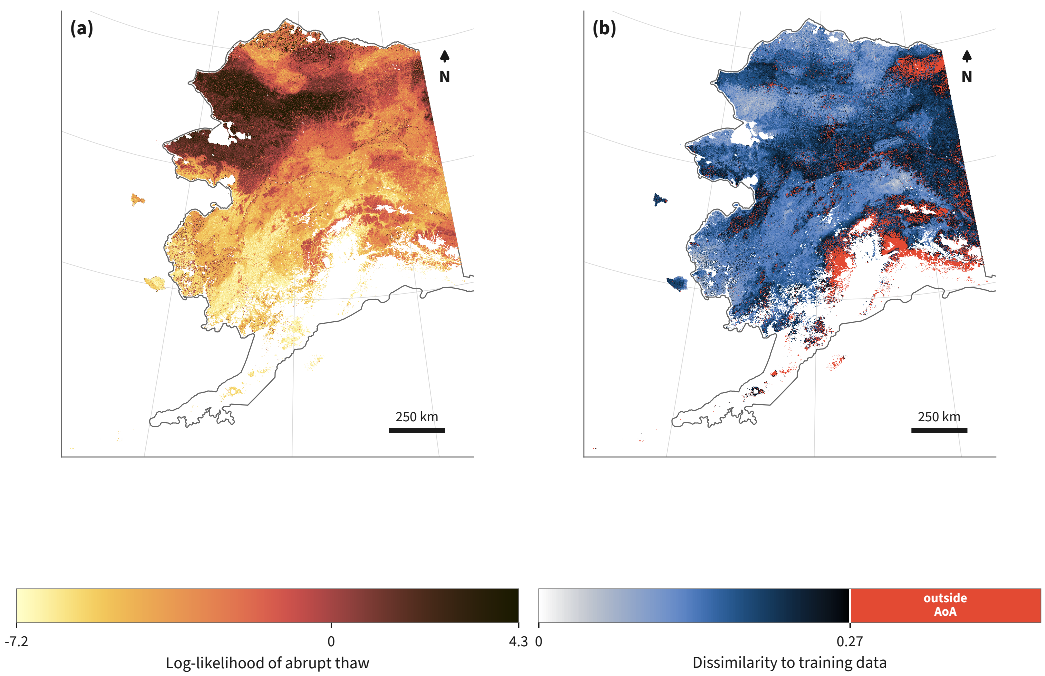

Figure 3. Statewide evidence of thaw mode and area-of-applicability. (a) Prior-free log-evidence index of thaw mode across Alaska's permafrost domain. Positive values correspond to environmental features that favor abrupt thaw, negative values similarly favor non-abrupt thaw, and zero log-evidence is considered neutral. The index characterizes similarity to observed thaw mode in the training dataset, not a probability of occurrence. (b) Dissimilarity between each grid cell and the nearest training observation in feature space, with features weighted by model-assigned importance, following the area-of-applicability (AoA) approach from Meyer and Pebesma (2021). Cells with a dissimilarity index at or below 0.27 fall within the AoA, while cells shaded in red represent extrapolation beyond the training data.

acteristics of the landscape, based on the input features and our model's interpretation, rather than a deterministic prediction of where abrupt thaw has occurred, is occurring, or will occur across the state.

Most of the domain is more consistent with non-abrupt thaw than abrupt thaw. The median in-AoA log-evidence is -2.46, so at the typical grid cell, the model favors the non-abrupt thaw mode by roughly a factor of 10. However, thaw mode is not evenly spread across the state. Abrupt-favored terrain concentrates in the tundra lowlands of western and northern Alaska, including the Seward Peninsula, the Brooks Range foothills, and the Arctic coastal plain. Conversely, the boreal interior and southern mountains primarily favor non-abrupt thaw. We investigate the spatial patterns of thaw mode in more detail in §4.3.

Figure 3b shows the dissimilarity index, which we use to determine the AoA for our model. Given the threshold set in §3.3, the mask flags 9.7% of the mapped domain as extrapolation beyond the envelope of the training features. The out-of-AoA cells are mostly due to features with long-tailed distributions, such as snow water equivalent (mean and trend), which accounts for nearly half of the dissimilarity score for flagged cells. Over the range we can measure, the model skill does not degrade with dissimilarity, so the bound of AoA shows where the input features are no longer represented by the training data, rather than where the model begins to fail.

4.2 Evaluating out-of-sample skill

Producing a trustworthy map of thaw mode depends entirely on the deployed model's ability to generalize outside of its training sample. For this task, our training dataset is not representative of the state of Alaska as a whole. Figure 4 compares the distribution of seven features between the training dataset and the in-AoA grid. These features were selected by hand to illustrate several known biases in the Alaska Permafrost Thaw Database, in addition to the geographic clustering shown in Figure 2 (Webb et al., 2026). Training points are flatter (median slope of 0.74 versus 3.4 degrees), closer to the nearest drainage (median height of 1 versus 15 m), and overall lower elevation (median height of 61 versus 260 m) than the rest of the state. They are much more likely to be open water (43% versus 3.8%) or emergent herbaceous wetlands (7.6% versus 3.9%), and much less likely to be deciduous forest (1% versus 3.7%), among other land cover indicators. This diver-

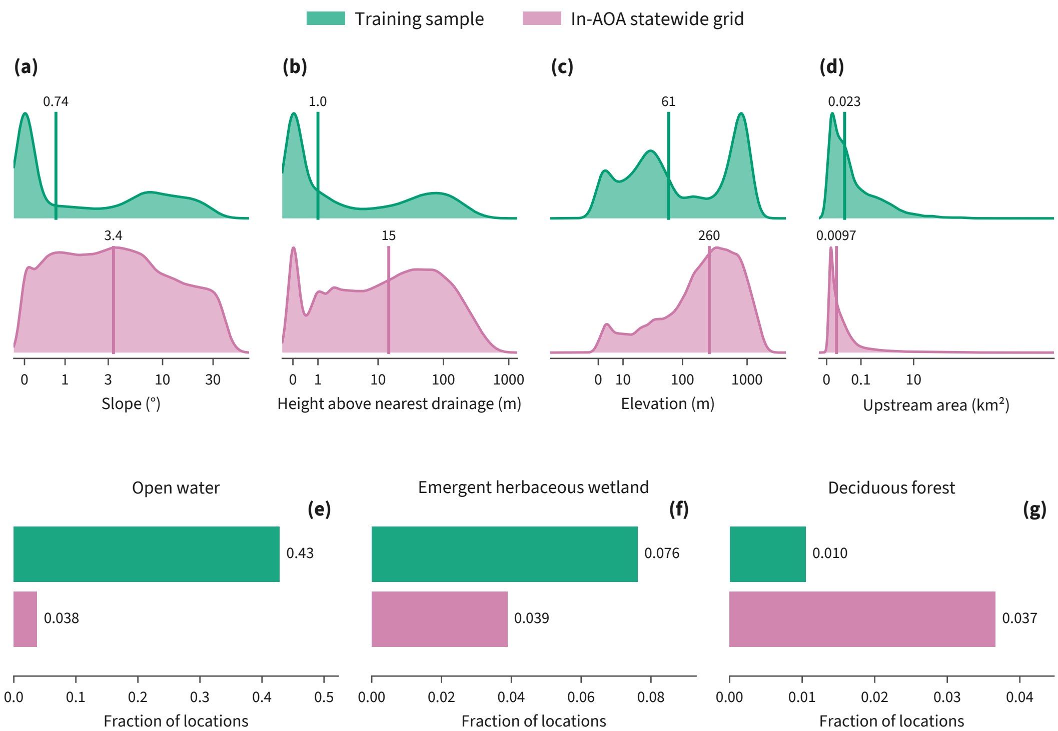

Figure 4. Representativeness of the training sample. Marginal distributions of selected environmental features in the training sample (green) and across the statewide grid within the area-of-applicability (pink). Continuous variables shown are slope (a), height above nearest drainage (b), elevation (c), and upstream drainage area (d). Densities are independently normalized, with vertical lines and annotations denoting the median of each respective distribution. Note that the horizontal axes use an inverse-hyperbolic-sine transform, approximately linear near zero and logarithmic at large values. Categorical variables shown are the land cover fraction of open water (e), emergent herbaceous wetlands (f), and deciduous forest (g).

gence motivates our decision to elevate area-of-applicability as a first-class model output alongside our log-evidence index, as well as a need for careful spatial cross-validation.

To evaluate how effectively the model can generalize outside of this geographically biased sample, we use spatial cross-validation, which holds out entire blocks of terrain so that nearby points cannot straddle the test-train splits. Figure 5a shows the full precision-recall curve for the operative 10 km block size, pooled across all splits, testing out-of-sample model skill. The model reaches an area under the precision-recall curve (AUC-PR) of 0.852±0.011 across 20 reshuffles of the block assignments, which is approximately fifteen times greater than the floor associated with random chance. The area under the receiver operating characteristic (ROC) curve, which measures false positives against true positives, is 0.98 over the same folds. However, because ROC is insensitive to the rare class, we do not use it as our primary metric.

Our baseline regression model recovers most of this performance. Logistic regression on the same folds achieves an AUC-PR of 0.776±0.017 (Figure 5a). We view this as the metric floor that a model must clear in order to provide useful predictive capability. The gradient-boosted model outperforms logistic regression across the full recall range, by a margin of 0.076 ± 0.019, stable across reshuffles.

Because the training points are concentrated regionally, we further investigate the model's robustness across space. Figure 5b tracks the AUC-PR against how far away the nearest training point typically lies from a given prediction. First, we construct block sizes of 5 to 200 km, corresponding to a median distance to the nearest training point of 1.8 to 30.8 km. To test larger distances, we cluster points using a k-means clustering algorithm on location only, producing between 3 and 50 larger approximate geographic regions, and perform the same hold-out analysis. The two methods overlap in the 20 to 30 km operational range, indicating an underlying pattern in model skill that is relatively insensitive to our choice in hold-out method. Skill decays gently from 0.88 when generalizing at a few kilometers to 0.54 at a median distance of 251 km. This AUC-PR would indicate a poor classifier on a balanced problem, but for our dataset is still roughly nine times greater than the chance floor. The predictions therefore survive extrapolation across hundreds of kilometers, rather than relying on a model that memorizes the feature space of specific neighborhoods.

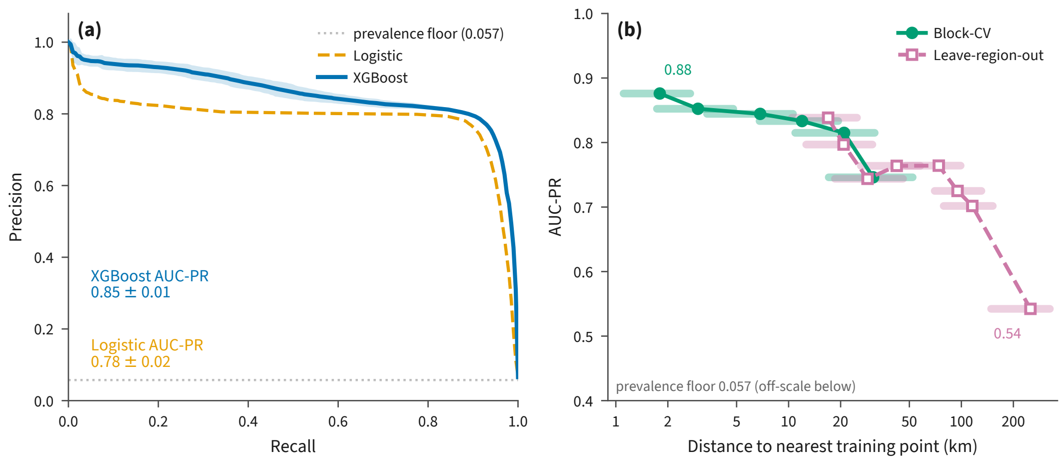

Figure 5. Model performance under spatial hold-out. (a) Pooled out-of-fold precision-recall curves for XGBoost and logistic regression models under cross-validation, using 10-km blocks. Curves show the mean across 20 shuffles of the block assignments, the blue ribbon denotes ±1 standard deviation for XGBoost, the dotted line marks the prevalence floor of 0.057, and annotations report the mean and standard deviation of the area under the precision-recall curve (AUC-PR). (b) XGBoost AUC-PR plotted against the distance from each observation to its nearest training point under block cross-validation (green) and leave-region-out cross-validation (pink). Markers indicate the median distance to the nearest training point for each holdout configuration, horizontal bars span the corresponding interquartile range, and the distance axis is logarithmic. Non-abrupt thaw is the positive class in both panels.

Additional controls confirm the signal is robust rather than an artifact of our software pipeline. Randomly permuting the training labels collapses skill to chance, giving an AUC-PR of 0.062 against the 0.057 floor. Trivial baseline models score at the floor. Only four pairs of training points share identical features while disagreeing on their label, so label noise imparts no meaningful ceiling on separation. Longitude and latitude are excluded from the feature table, so the model does not ever use location directly as a predictor.

4.3 Identifying key indicators of abrupt thaw

Rather than attempt to attribute importance across 70 variables, many of which are themselves interrelated, we group the model's features into 22 emergent families. We define these families by their correlation in feature space, before computing any SHAP values, so that the groupings remain independent of the importance attribution. Figure 6 ranks these families by their combined importance, that is, the mean absolute sum of SHAP values of each of their members. Note that importance does not yet tell us which thaw mode is favored, but simply the overall magnitude of that family's contribution to the model's classification. Alpine relief leads at 23%, followed by annual and dry season temperature at 16%, land cover at 12%, and thermal continentality at 9%. The trailing 18 groupings each contribute under 7% of the total influence on the model.

At the coarser domain level, no single control emerges as a dominant driver of the model's predictions. Importance between different groupings cannot be neatly summed together, such as alpine relief and surface curvature, so we compose a second level of thematic groupings, based on domain knowledge. Topography leads at 24.6% of the domain-aggregated importance, with baseline temperature close behind at 20.7%. Snow water equivalent (including mean and trend) and the combined land cover and fire history variables each contribute 14.2% of the importance, with non-snow precipitation rounding out the top five with 9.1%. The remaining thematic groupings, namely climate seasonality, soil properties, summer temperature, and yedoma, each contribute under 7%.

To examine the direction that features push the model, we turn from aggregate groupings to look at the nine most impactful individual variables. Figure 7 plots each of those variables' SHAP values against the actual value of the variable itself. Positive SHAP values indicate that the model was pushed towards abrupt thaw, while negative values in-

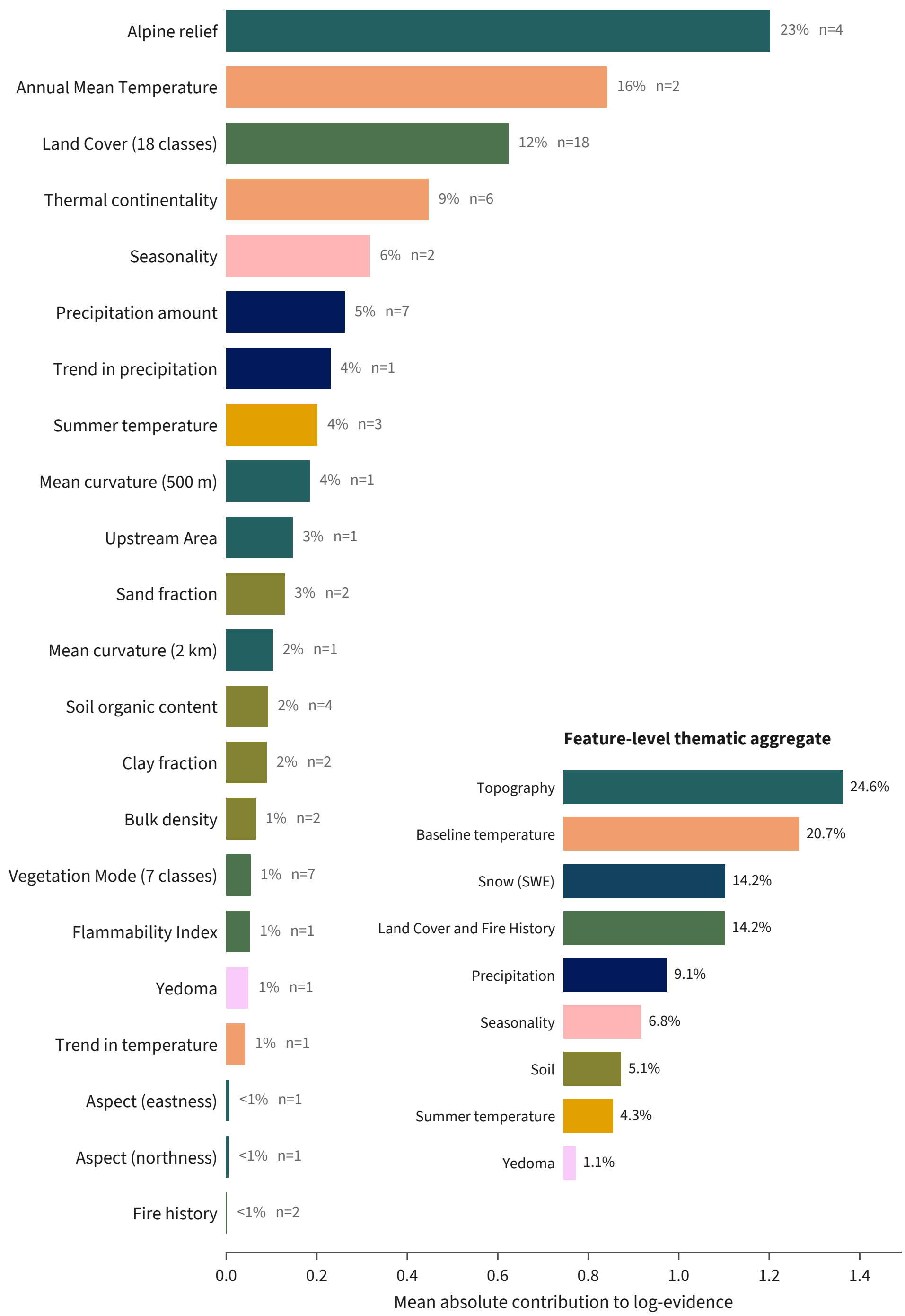

Figure 6. Grouped SHAP importance of environmental variables. Environmental variables are organized into 22 feature families based on their correlation structure and ranked by their SHAP importance on held-out observations under spatial cross-validation. Mean contribution to log-evidence is calculated by summing the SHAP values of member variables and taking the absolute value. Inset shows SHAP importance aggregated across thematic domains, denoted also by the color of bars on both plots, with percentages showing the share of total importance.

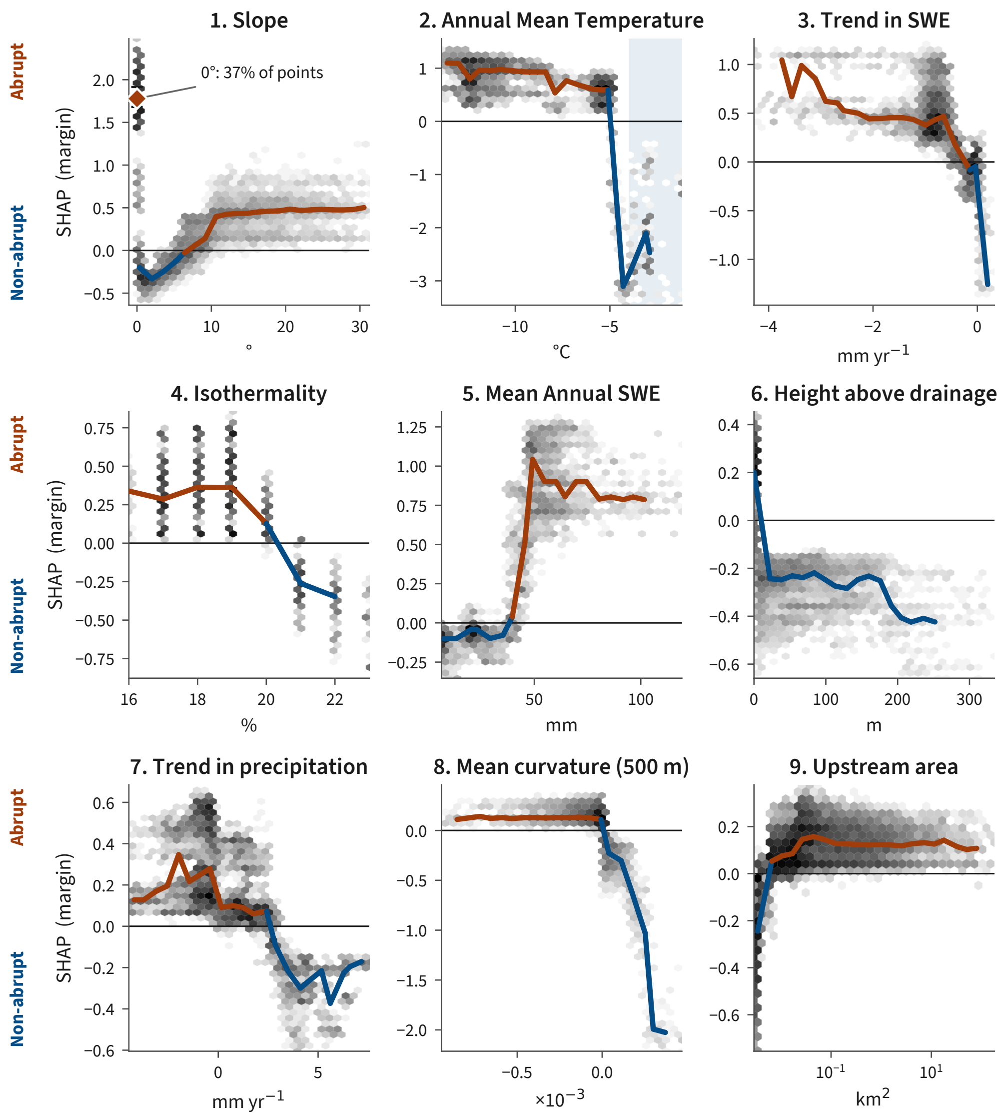

Figure 7. Individual SHAP response plots for leading indicators. SHAP values for the nine most influential continuous variables on held-out observations, ranked by mean absolute importance, ordered from top-left to bottom-right. Grey shading shows observation density, while binned median responses show where each variable favors abrupt (orange, positive y) or non-abrupt (blue, negative y) thaw. Note that vertical and horizontal scales vary among panels.

dicate that the model was pushed towards non-abrupt thaw. The topographic variables share a clear story, in which abrupt thaw is favored where slopes are near 0°, height above nearest drainage is lowest, mean curvature is negative, and there is greater upstream drainage area. In the training data, 37% of points sit at a slope of 0°, almost always associated with open water features. Higher slope values, above roughly 10°, also favor abrupt thaw, indicating a bimodal contribution to thaw mode from slope alone. The climatic variables also share some similarities, with abrupt thaw favored by the model where annual mean temperatures are cold and mean annual snowpack is high, corresponding also to places where the trend in snow water equivalent is declining more rapidly, and isothermality and trends in overall precipitation are low.

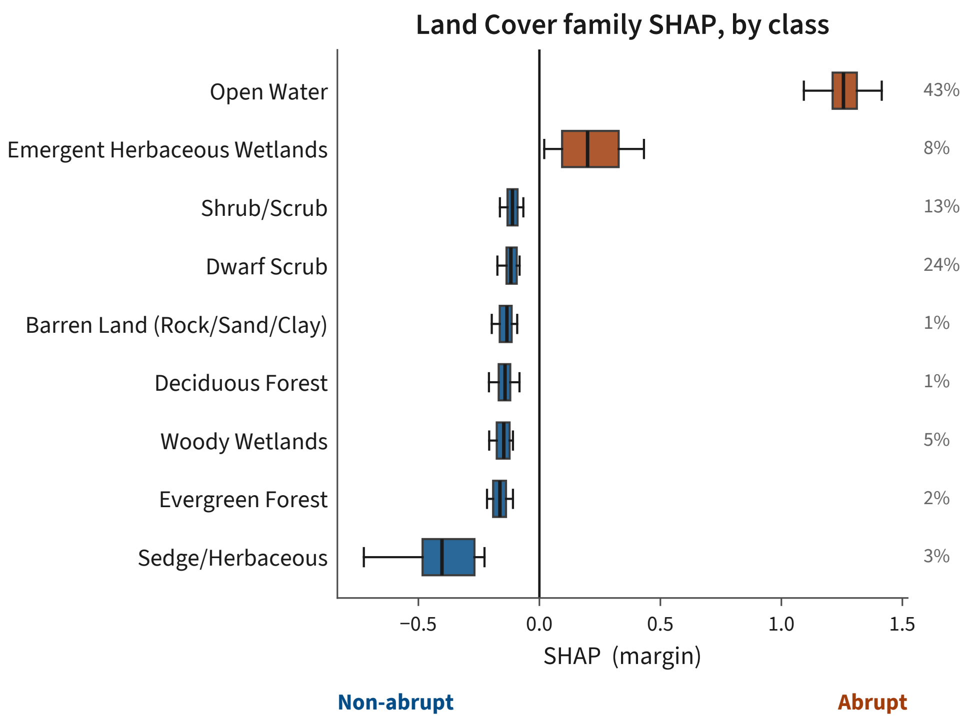

Figure 8. Land-cover SHAP contributions. Distributions land-cover family SHAP values under held-out observations for classes with at least 100 points. Boxes show the median and interquartile range, whiskers span the 5th-95th percentiles, and labels give each class's share of observations. Positive values (orange) favor abrupt thaw and negative values (blue) favor non-abrupt thaw.

Land cover is encoded as one-hot columns, so we look at its contribution to the model on a class-by-class basis in Figure 8. It is also the family with the strongest sign change in SHAP values across different categories. "Open Water" and "Emergent Herbaceous Wetlands" push the model towards abrupt thaw, while all other landcover classes push the model towards non-abrupt thaw. "Sedge/Herbaceous" land cover provides the strongest directional push towards non-abrupt thaw of those classes. Note, however, that the model still classifies some number of abrupt and non-abrupt cells within all of these land cover categories.

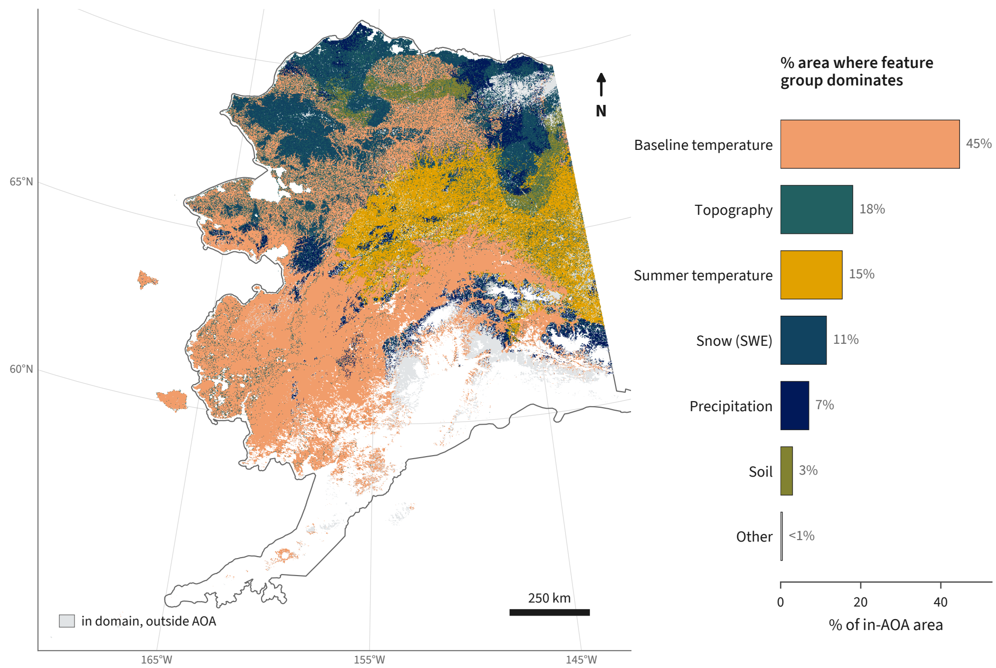

Figure 9. Spatial distribution of dominant indicators. Each in-AoA grid cell is colored by the thematic feature group with the largest-magnitude net SHAP contribution from the operational model. Bars show the percentage of in-AoA area dominated by each group. Grey shading marks terrain within the permafrost domain but outside AoA.

Finally, in Figure 9, we return to the susceptibility map to look at how feature importance is distributed in space. For every in-AoA grid cell, we identify which thematic grouping (Figure 6, inset) exerts the greatest influence on the model's prediction in that grid cell. Note that this is distinct from summed importance across the domain, because it is a winner-takes-all metric in each individual cell. We find that no single theme wins everywhere across the state. Baseline temperature is the most frequent driver, covering

45% of grid cells, especially prevalent in the southern and western regions. Topography governs 18% of grid cells, especially where open water features are found, and in the foothills of the Brooks Range and across the North Slope. Summer temperature dominates in 15% of the grid, concentrated in the continental interior. The other groupings show up sporadically, each giving a majority influence on the model's classification in less than 12% of the in-AoA domain. The regional structure of the SHAP importance does not follow smooth spatial trends in the underlying data products, but instead switches between key indicator groups as the local terrain and climate vary.

5 Discussion

We produce a continuous, predictive map of thaw mode across Alaska's permafrost domain. The map is a characterization of the landscape, rather than a forecast: a positive log-evidence value means that the local features resemble settings where abrupt thaw has been observed, but is not a deterministic prediction of future events. Predisposition is continuous, so there is no particular threshold that flips terrain from "abrupt" to "non-abrupt." The median grid cell in our area-of-applicability (AoA) has a value of -2.46, corresponding to non-abrupt thaw, consistent with our prior understanding that abrupt thaw is only likely to impact a minority of the permafrost region. Roughly 90% of Alaska's permafrost domain falls within the AoA of our model, and we do not report any results outside that AoA. We choose zero as a convenient log-evidence threshold for reporting thaw mode fraction, which is conservative, given that a grid cell with log-evidence of zero is more than ten times more likely associated with abrupt thaw than the median of the landscape. Under this threshold, we find that 24.7% of the in-AoA permafrost region favors abrupt thaw. This finding is comparable to previous work that has estimated 20% of the northern permafrost region is susceptible to abrupt thaw on the basis of topography and ground ice mapping (Olefeldt et al., 2016; Turetsky et al., 2020).

Mechanistically, our model makes its strongest distinctions on topography and temperature, both well-recognized controls on abrupt thaw. This is reassuring, as it indicates that the classifier is reconstructing known physical processes, rather than identifying spurious or incidental relationships in the labels or features. Temperature sets a regional envelope, where cold regions favor the development of large ground ice reservoirs, while warmer regions tend not to be able to sustain the excess ground ice required for abrupt thaw to proceed. In Figure 9, we see this effect, where baseline temperature dominates 45% of the in-AoA grid cells, despite only contributing 20.7% of the domain importance in Figure 6, inset. Topography has a much more distinct structure in the model, displaying a bimodal indication of abrupt thaw. In terrain that is low-lying, close in elevation to its nearest drainage, concave, and gathering water from a large upstream drainage area, the model finds a strong signal of abrupt thaw. This is consistent with our understanding of classic thermokarst landscapes, and is further reinforced by the importance the classifier places on the Open Water and Emergent Herbaceous Wetlands land cover classes. At the opposite extreme, on slopes steeper than roughly 10°, abrupt thaw becomes favored again, now in terrain characterized by retrogressive thaw slumps and active layer detachments. This is further supported by the importance assigned to mean annual snow water equivalent (SWE), since snowpack accumulates orographically on upland terrain. Approximately 30% of our abrupt-favoring cells are in upland environments, although this result should be viewed carefully, given how many of our abrupt thaw observations in the training dataset come from low-lying wetlands. Overall, we find that the model is using temperature and topography to bound the overall envelope and distinguish between distinct pathways of the abrupt thaw mode.

Beyond topography and temperature, the model draws on a second tier of climate descriptors, which it uses to refine the thermal envelope discussed above. The cold-season temperature variables track annual mean temperature closely and merge into the baseline temperature signal in Figure 9. Isothermality is the fifth strongest individual feature, with increasing isothermality correlated more strongly with non-abrupt thaw (Figure 7). However, the seasonality thematic grouping, which includes isothermality, only controls 0.16% of the in-AoA cells in Figure 9, indicating a broader, more diffuse contribution that is rarely the deciding factor in any given grid cell. The trend in snow water equivalent (SWE) escapes some of this climate-related ambiguity, emerging as the fourth strongest individual feature in the model, behind only slope, mean annual air temperature, and open water. Declining snowpack favors abrupt thaw (Figure 7), a direction echoed by declining precipitation. Unlike the descriptors above, trend in SWE is nearly independent of elevation and only modestly tied to mean temperature in our feature space. These features help distinguish abrupt thaw from non-abrupt thaw in a classification task, even if their mechanistic role in the process is not clear.

The permafrost community heavily associates several drivers with abrupt thaw that end up carrying far less weight in our model than expected. We interpret this as a statement of the model architecture and underlying data products, not as a revision of our established process understanding. For some variables, we find a real signal, but one that is redundant with other features that are more clear in the model. On their own, nitrogen content and soil organic carbon separate abrupt from non-abrupt thaw sites nearly as well as annual mean temperature, and the climate-derived flammability index also performs relatively well. Yet all of them nearly vanish in the fitted model, where soil properties together contribute only 5.1% of the domain importance and flammability only 1%. This indicates that the gradient-boosted tree is routing around these features, preferring slope and elevation over the collinear soil properties, and broader climate variables over the derived flammability index. For a second group of variables, the data products themselves do not carry enough information for the model to learn the underlying mechanism. On its own, yedoma extent barely separates the two thaw modes better than chance, despite the known mechanistic importance of ground ice. This adds to a growing body of evidence that our yedoma maps are not aligned with where abrupt thaw is observed on the landscape, as discussed further in Webb et al. (2026). Similarly, while we recognize that abrupt thaw is often triggered by disturbance, including wildfire (Jones et al., 2015), our burn count and time-since-last-fire features barely differentiate the thaw modes above baseline. In this case, it is likely that the timing matters, especially relative to when thaw mode is observed, and this relationship never enters a static feature stack like the one our model learns on here.

Our interpretations of the model-assigned feature importance rest on the underlying assumption that the model has learned something about the underlying landscape, rather than the particular training sample. A natural concern is that the classifier has simply memorized the sampling bias from the Thaw Database, such that our map of thaw mode traces where samples were collected rather than where abrupt thaw is actually favored. Two key results argue against this reading: generalizability and the spatial distribution of key SHAP families. First, the model generalizes far beyond its training points. When we hold out entire regions, forcing the model to extrapolate far away from where it was trained, skill decays to an AUC-PR of 0.54 at a median distance of 251 km, roughly nine times greater than the chance floor. A model that had memorized the sampling footprint but not learned any underlying pattern in the data would collapse towards an AUC-PR of 0.057 in this limit. Second, the dominant driver is not fixed spatially across the state, and not even within the locations with the highest density of training samples. In

Figure 9, the leading indicator switches between baseline temperature, topography, and other thematic groupings as the local variables change, rather than tracing a single proxy for sample location everywhere. We do not claim that bias and mechanism are fully separable, in part because the low-lying wetlands that are oversampled in the training data are also where thermokarst features genuinely concentrate. We therefore rely on the generalization and spatial variation tests above, rather than attempting to statistically remove the sampling geography.

This inseparability between bias and mechanism is the first of several real limits on how far we can push our interpretation of the statewide map. The primary limitation is that our analysis is fully static. The log-evidence map describes the present predisposition of the landscape, but cannot differentiate between where abrupt thaw has already occurred, where it is currently underway, and where it will proceed in the future. We deliberately strip dates of imagery acquisition from our training points and aggregate through time where necessary in our source data products, because our training labels are static in time, and so carry no temporal information about when abrupt thaw was triggered. We produce a snapshot of the landscape as it is today, rather than a trajectory through time. A second limitation is that we lack independent, ground-truth data to validate our model's classification. Every evaluation in this study depends on held-out points from the dataset itself, rather than a true independent verification from a different data product. Finally, we stress again that the indicators we identify are useful for distinguishing thaw mode, but cannot be assigned any causal role by this model. As such, we lean on a mechanistic explanation only where established prior work supports it. With these bounds in mind, we turn to a broader discussion of what our results mean for understanding thaw mode statewide.

We find that 24.7% of Alaska's in-AoA permafrost region favors abrupt thaw over non-abrupt thaw, but that percentage hides a landscape that is sharply partitioned. In Figure 10, we aggregate our thaw mode results by physiographic grouping and ecoregion. Roughly two-thirds of grid cells in the Brooks Range and the Seward Peninsula are more consistent with abrupt thaw than non-abrupt thaw, followed by the Arctic Foothills at 47%. The five regions with the greatest proportion of grid cells favoring abrupt thaw are all tundra, while the signal is more subdued in the taiga, and all but disappears in the forested mountains of the interior and the south. By percentage of total area, tundra regions are more susceptible to abrupt thaw than would be expected by our statewide

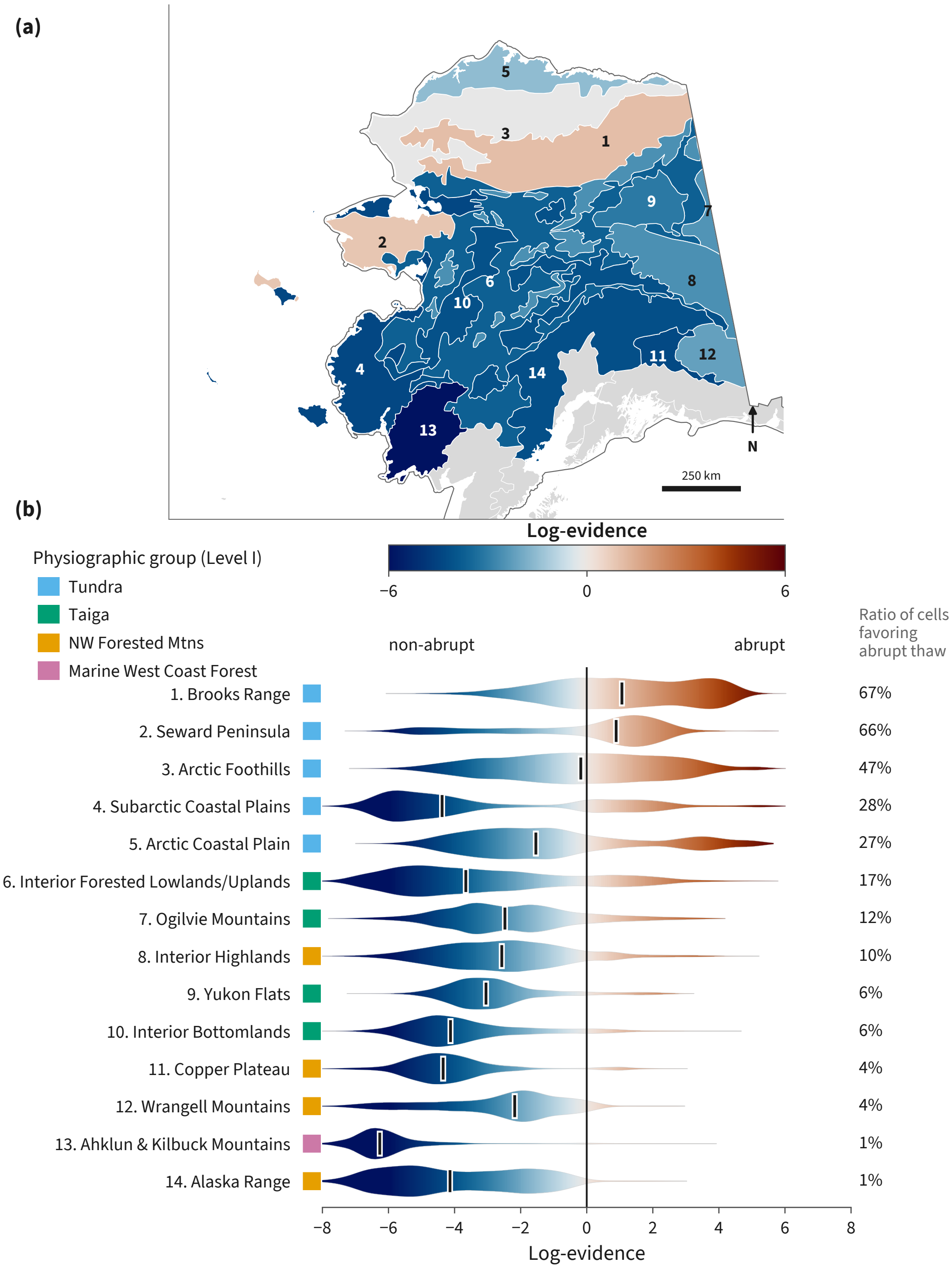

Figure 10. Regional variation in thaw mode. (a) Median log-evidence by Level III ecoregion. (b) Distributions of in-AoA log-evidence within each ecoregion, ranked by the percentage favoring abrupt thaw. Ticks mark medians and colored squares denote Level I physiographic groups. Only regions with at least 50% permafrost coverage are shown, and all retained regions have at least 80% AoA coverage.

ratio, while taiga and forested regions are much less. The spread of log-evidence within each region is wide, and in most cases wider than the differences between regional medians. Physiography therefore only organizes thaw mode at a coarse scale. An ecoregion sets many of the environmental variables that landscape scale predictions of thaw mode depend on, while our continuous log-evidence index displays fine-scale structure that is not captured by a categorical map.

The categorical map most relevant for abrupt thaw is the thermokarst landscape classification from Olefeldt et al. (2016). We do not position this comparison as a validation step, since the two products have different aims, but instead examine how the classes from Olefeldt et al. (2016) cut across our characterization of thaw mode. Figure 11 shows that thaw mode (this study) is nearly orthogonal to thermokarst landform and occurrence potential (Olefeldt et al., 2016). Potential for thermokarst development (ranked from "low" to "very high") explains 7% of the variance in our prediction of thaw mode in hillslope thermokarst, 3% in lake thermokarst, and 1% in wetland thermokarst. The only identifiable trend occurs in hillslope terrain, which does lean further towards abrupt thaw in our model as the occurrence potential rises.

The two products are not built to agree, and their poor correlation reflects a difference in how they treat the underlying reality of the landscape that they map, rather than any methodological shortcoming. The Olefeldt et al. (2016) classification investigates thaw occurrence, ranking where thermokarst landforms of a given family are present or could potentially develop from ground ice and topography. Our index investigates thaw mode, asking which pathway the landscape favors once thaw is underway. Webb et al. (2025) define these two axes as distinct, by definition, since abrupt thaw is broader than thermokarst and is set by the rate and ecosystem impact of thaw rather than by ground ice content or geomorphology alone. A map of thermokarst occurrence therefore cannot stand in for a map of thaw mode, and vice versa. The correlation that does remain between the two products strengthens as we move from wetland, to lake, to hillslope thermokarst, which mirrors the order of these families from the slowest expression of landform development to the fastest (Webb et al., 2025). Where the Olefeldt et al. (2016) landscape is built from retrogressive thaw slumps and active layer detachments, our thaw mode index and their occurrence potential start to converge.

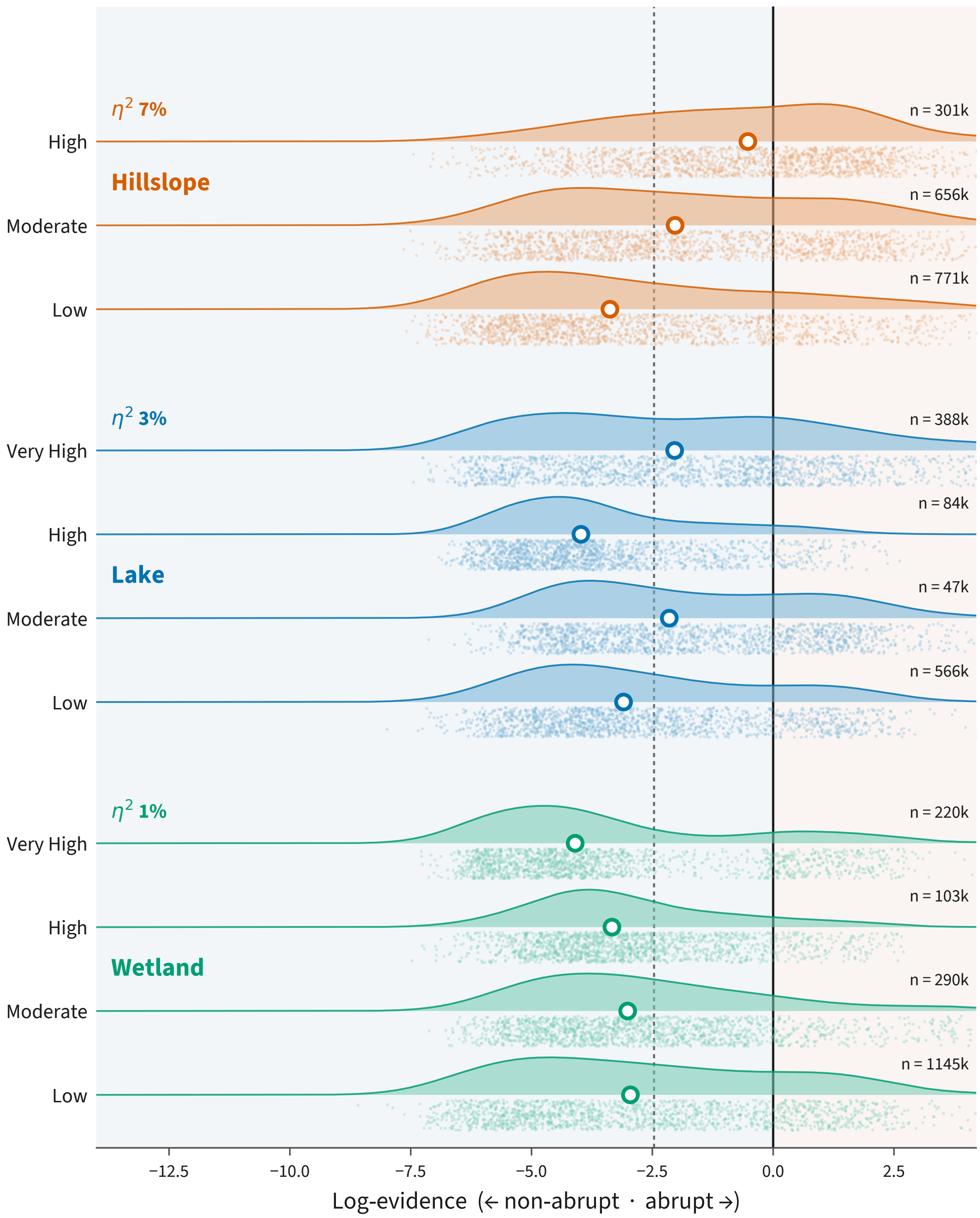

Figure 11. Thaw mode across thermokarst landform classes. Distributions of in-AoA log-evidence by thermokarst type and occurrence potential from Olefeldt et al. (2016). Curves show density, circles mark medians, solid and dashed vertical lines mark neutral log-evidence and the statewide median, respectively. Points sub-sample grid cells. Annotated η^2 values give the variance in log-evidence explained by occurrence potential within each thermokarst type, and n gives the number of in-AoA grid cells. "None" classes are excluded.

Across Alaska's permafrost domain, the distinction between abrupt and non-abrupt thaw has until now only been defined by scattered, expert-labeled points. Existing statewide products resolve where permafrost occurs, where specific landforms develop, or where terrain could thaw, but none tell us which thaw pathway an arbitrary, unvisited site favors. By turning roughly $19,000$ labeled points into a continuous log-evidence surface with a paired reliability bound, we can now pose that question across the domain. Rather than focusing on a single areal fraction, we can highlight regions where abrupt thaw is favored, such as the Brooks Range or the Seward Peninsula, or where non-abrupt thaw is favored, such as the interior mountains. We can get increasingly specific, up to the resolution of the prediction grid, which is approximately 1 km^2 here. For infrastructure and community risk, the broad regional structure is a screening step, allowing stakeholders to prioritize finer hazard assessment and monitoring where they are most warranted.

Abrupt thaw mobilizes carbon more rapidly, from deeper stocks, and often as preferentially more methane than non-abrupt thaw. Previous syntheses have established that abrupt thaw contributes proportionally more to the permafrost-carbon feedback than its area fraction would suggest (Olefeldt et al., 2016; Walter Anthony et al., 2018; Turetsky et al., 2020). Yet, Earth System Models that predict future permafrost thaw rarely include the abrupt thaw mode, or do so by representing its spatial footprint through aggregate inventories of landform features (Nitzbon et al., 2020). Our map of thaw mode provides a continuous, statewide field of where the landscape favors the abrupt pathway, so that abrupt thaw can be conditioned spatially at any site, rather than lumped into a domain-wide total or tied to a specific process. This is a spatial layer that previous inventories and risk mapping products have lacked, identifying where on the landscape abrupt and non-abrupt thaw processes can be disentangled for quantitative carbon modeling.

As the map can guide hazard screening and carbon modeling, so too can it guide the observations needed to improve itself. The pairing of a log-evidence surface with an explicit reliability bound hands us a tool for targeting the next most informative observations. The terrain that falls outside our model's AoA marks where the model has no basis for prediction, and thus where we need expert-verified labels of thaw mode. The Arctic Foothills hold the second-largest share of abrupt-favoring terrain in the state, yet are sampled roughly three times less densely than the adjacent Brooks Range. The Interior Forest Lowlands and Uplands form the single largest ecoregion in the domain and contribute the third-largest share of abrupt-favoring terrain, but contain fewer than 200 labeled training points. Directing field campaigns towards these under-sampled regions and focusing on specific site profiles that lie outside our AoA would provide the sharpest improvement for the current map.

Thaw mode predictions can be further improved by strengthening the underlying feature stack that the model uses to learn about the landscape. Higher spatial and temporal resolution datasets of fire history, subsurface properties, and specifically yedoma extent could let a future classifier recover the mechanisms that it currently routes around, promoting drivers that are hidden in the data today into leading indicators. The greatest challenge, however, is to move from a static snapshot of the landscape features as they appear today to a dynamic trajectory of thaw mode as it progresses. Our map describes the predisposition of Alaska's permafrost terrain to the abrupt or non-abrupt thaw mode, but not the timing or rate at which thaw will proceed. Adding temporal information is far beyond the scope of this work, but is the next critical step towards forecasting risk from abrupt thaw, including direct impacts on human communities and infrastructure and climate impacts through the permafrost-carbon feedback.

6 Conclusion

We present the first continuous, statewide map of thaw mode across Alaska's permafrost domain. By combining $19,288$ expert-labeled observations with 70 environmental indicators, we produce a prior-free log-evidence index showing whether points on the landscape more closely resemble abrupt or non-abrupt thaw, paired with an explicit area-of-applicability (AoA) bound. Within that area, which covers roughly 90% of the permafrost domain, we find 24.7% of the landscape has features more consistent with abrupt thaw than non-abrupt thaw. This fraction describes present predisposition, not the observed extent, probability, or future occurrence of abrupt thaw. The classifier achieves an AUC-PR of 0.852, approximately fifteen times greater than the chance floor, and retains an AUC-PR of 0.54 even when tested on observations at a median of 251 km away from the nearest training point.

Topography and temperature provide the leading distinction between thaw modes, but their influence varies spatially across Alaska. Flat, low-lying wetlands and steeper uplands both favor abrupt thaw, consistent with distinct pathways associated with common landforms like thermokarst lakes and retrogressive thaw slumps, respectively. No single indicator controls this signal statewide, but baseline temperature is the dominant influence on the model in 45% of in-AoA grid cells, with the rank importance of leading indicators shifting considerably in space as local environmental factors vary.

These local differences aggregate into a sharply partitioned statewide pattern. Roughly two-thirds of the Brooks Range and Seward Peninsula favor abrupt thaw, as does just under half of the Arctic Foothills, while non-abrupt thaw is favored across the forested interior and mountain regions. Thaw mode nevertheless varies widely within each region, retaining landscape-scale structure that physiographic categories and previously mapped thermokarst occurrence potential do not capture. Our map of thaw mode therefore resolves a largely distinct and complementary dimension of permafrost change. This work helps sharpen our estimates of thaw-related risks to Alaskan communities and where the permafrost-carbon feedback will draw most heavily on abrupt thaw under future climate change.

Open Research Statement

Statement text.

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Appendix A Data sources

Table A1. Geospatial data products used to build the feature set. "Cols" gives the number of columns each product contributes to the 70-column model matrix, after one-hot encoding of the categorical land-cover and vegetation-mode sources and depth aggregation of the soil variables.

Data product	Native res.	Variables derived	Cols	Reference
Remote sensing				
USGS 3DEP DEM	10 m	Elevation, slope, aspect (northness, eastness), mean curvature (500 m, 2 km)	6	(U.S. Geological Survey, 2019)
MERIT Hydro v1.0.1	~90 m	Height above nearest drainage, upstream area	2	(Yamazaki et al., 2019)
NLCD 2016 (Alaska)	30 m	Land cover (one-hot)	18	(Dewitz, 2019)
MODIS MCD64A1	~500 m	Time since last fire, burn count	2	(Giglio et al., 2018)
Community and modeled data products				
SoilGrids (ISRIC)	250 m	Organic carbon, nitrogen, bulk density, sand, clay (two depths)	10	(Poggio et al., 2021)
ALFRESCO (SNAP)	~1 km	Flammability, vegetation mode (one-hot)	8	(Bennett et al., 2017)
IRYP v2 (yedoma)	Vector	Confirmed yedoma presence	1	(Strauss et al., 2022)
Gridded climate				
WorldClim v1	~1 km	19 bioclimatic variables	19	(Hijmans et al., 2005)
Daymet V4	1 km	Mean annual SWE, and SWE, precipitation, and temperature trends (1991–2020)	4	(Thornton et al., 2022)
Total			70	

Appendix B Hyperparameters

We selected the XGBoost hyperparameters by grid search within the inner loop of the nested cross-validation of § 3.2. The outer loop holds out spatial blocks to estimate skill, and within each outer training set we run a second five-fold spatial split to score every candidate configuration. We use 10-km equal-area blocks with no buffer. The search grid crosses five hyperparameters for 32 combinations in total: maximum tree depth (3, 5), minimum child weight (5, 20), L2 regularization (1, 10), learning rate (0.05, 0.1), and number of trees (200, 400). For each candidate, we measure the area under the precision-recall curve on the held-out inner folds and keep the highest scoring configuration.

This grid is deliberately narrow. Earlier exploration on previous versions of the labeled dataset searched a wider space that included deeper trees and row and column subsampling, but we set that aside in favor of spatially-blocked evaluation, given how costly nested cross-validation is.

The selection is not identical across folds, but consistently favors shallow and strongly regularized trees with a slow learning rate. The final configuration has a maximum depth of 3, minimum child weight of 5, L2 penalty of 1, a learning rate of 0.05, and 200 trees. For the logistic regression baseline model, we only tune the inverse regularization strength over 0.01, 0.1, and 1.0 in the same nested loop.

Appendix C Dissimilarity index

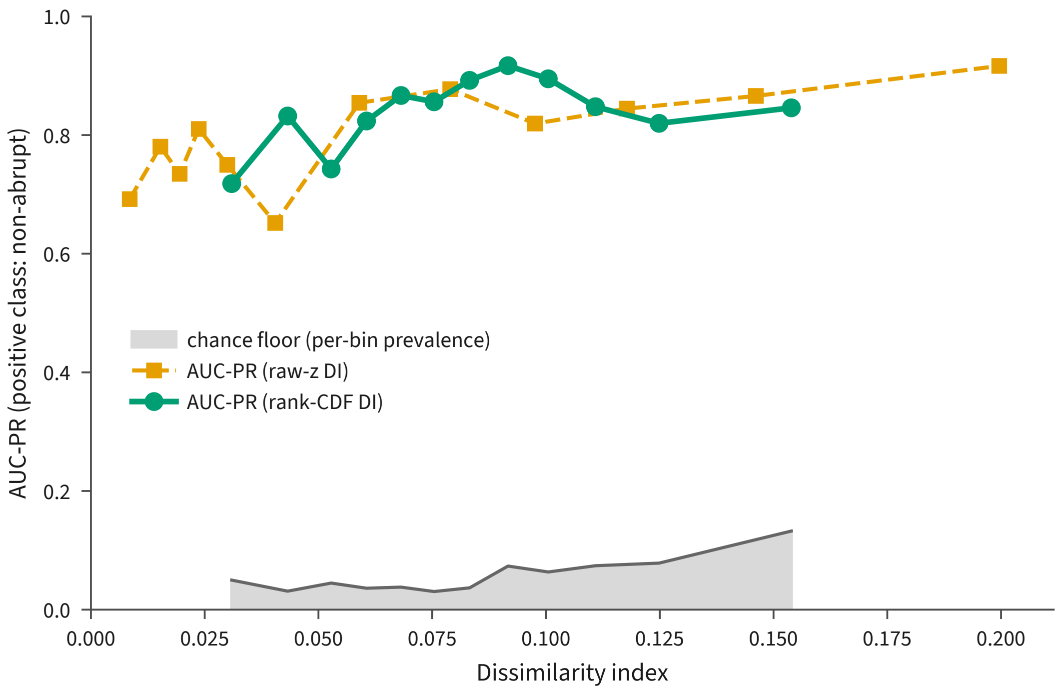

Figure C1. Model performance against dissimilarity index. Out-of-fold model skill does not degrade as dissimilarity in feature space rises. We plot pooled out-of-fold AUC-PR from the operative model, computed under spatial cross-validation and binned into twelve equal-count bins by dissimilarity index, using non-abrupt as the positive class. Points sit at each bin's median dissimilarity and the grey band is the per-bin prevalence floor. Across the sampled range, the rank-CDF AUC-PR stays between 0.72 and 0.92, well above the floor, and does not trend downward as dissimilarity grows. The green curve uses the operative rank-CDF dissimilarity index, while the orange curve repeats that analysis with the Meyer-Pebesma raw-z coordinate. Dissimilarity does still track the magnitude of the model's error, with a Spearman correlation of 0.49 between the rank-CDF index and the absolute out-of-fold residual.

Appendix D SHAP feature families

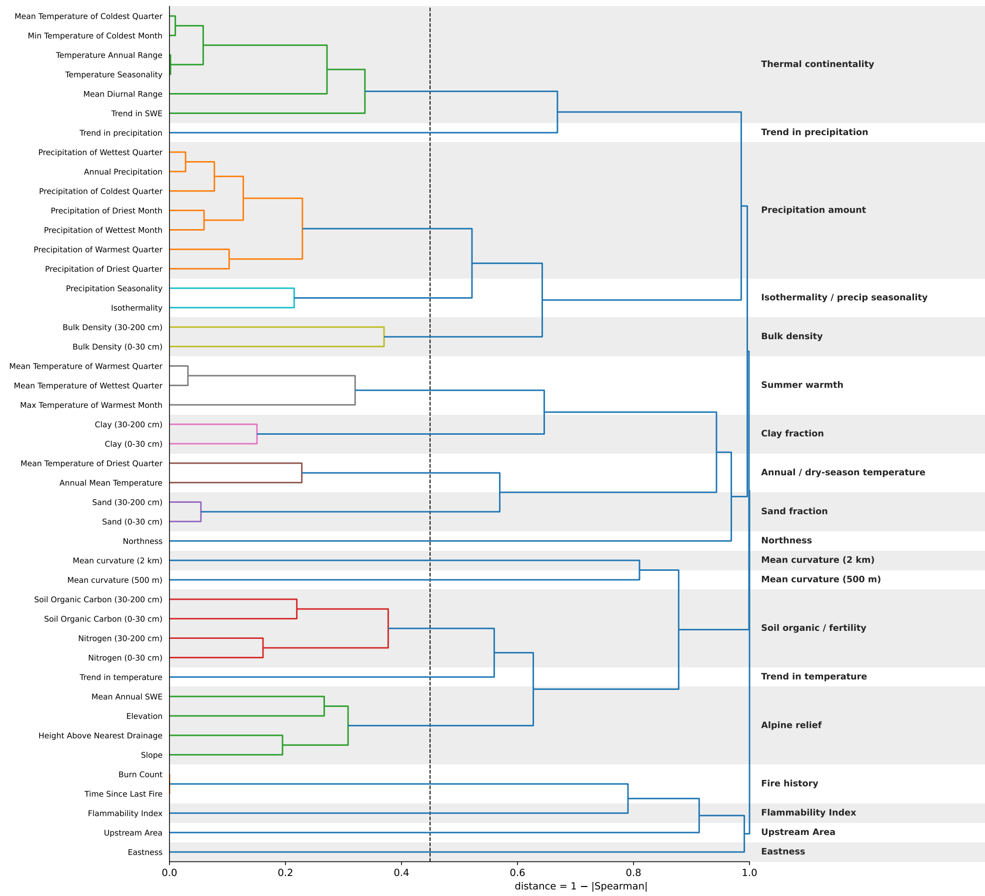

Figure D1. Emergent feature families from correlation structure. We group the 70 environmental variables into 22 families before computing any SHAP values, so that the groupings stay independent of the importance attribution. Continuous variables are clustered on feature-space rank correlation. The distance between two variables is 1 minus their absolute Spearman correlation, so perfectly correlated (or perfectly anti-correlated) variables sit at a distance of 0 from one another. Clusters are merged by complete linkage, meaning that two clusters join on the distance of their farthest members. The number of families is not fixed in advance: we cut the tree at the midpoint of the largest gap in merge heights, marked by the dashed line at a distance of 0.449. The absolute correlation within every family is thus at or above 0.55. Each shaded band delimits one emergent family and carries the name used in Figure 6. Categorical variables are not included here, but are collapsed to one family per data source, yielding three additional families for land cover, vegetation mode, and yedoma extent.